IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2019

WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

1. (a) We consider the sequence $p[k] = \langle g[n], g[n-k] \rangle$. The goal is to find $P(z)$ first then the z-transform of $\langle g[n], g[n-3k] \rangle$ is obtained by downsampling $p[k]$ by 3 and expressing that in the z-domain

[2/6]

Using the definition of the z-transform and of the inner product we have:

$$P(z) = \sum_k p[k] z^{-k} = \sum_k \sum_n g[n] g[n-k] z^{-k}$$

Replacing $n - k = j$ yields

$$P(z) = \sum_n \sum_j g[n] g[j] z^{j-n} = G(z) G(z^{-1}).$$

[4/6]

We finally downsample by 3 and obtain

$$\langle g[n], g[n-3k] \rangle = \delta_k \Leftrightarrow \sum_{k=1}^{3} G(W_3^k z^{1/3}) G(W_3^k z^{-1/3}) = 3.$$

[6/6]

(b) i.

$$Y(z) = G_0(z) X_0(z^3) + G_1(z) X_1(z^3) + G_2(z) X_2(z^3)$$

[2/8]

Consequently

$$\hat{X}_0(z) = \frac{X_0(z)}{3} \left[\sum_{k=1}^{3} H_0(W_3^k z^{1/3}) G_0(W_3^k z^{1/3}) \right] + \frac{X_1(z)}{3} \left[\sum_{k=1}^{3} H_0(W_3^k z^{1/3}) G_1(W_3^k z^{1/3}) \right]$$

$$+ \frac{X_2(z)}{3} \left[\sum_{k=1}^{3} H_0(W_3^k z^{1/3}) G_2(W_3^k z^{1/3}) \right];$$

$$\hat{X}_1(z) = \frac{X_0(z)}{3} \left[\sum_{k=1}^{3} H_1(W_3^k z^{1/3}) G_0(W_3^k z^{1/3}) \right] + \frac{X_1(z)}{3} \left[\sum_{k=1}^{3} H_1(W_3^k z^{1/3}) G_1(W_3^k z^{1/3}) \right]$$

$$+ \frac{X_2(z)}{3} \left[\sum_{k=1}^{3} H_1(W_3^k z^{1/3}) G_2(W_3^k z^{1/3}) \right]$$

and

$$\hat{X}_2(z) = \frac{X_0(z)}{3} \left[\sum_{k=1}^{3} H_2(W_3^k z^{1/3}) G_0(W_3^k z^{1/3}) \right] + \frac{X_1(z)}{3} \left[\sum_{k=1}^{3} H_2(W_3^k z^{1/3}) G_1(W_3^k z^{1/3}) \right]$$

$$+ \frac{X_2(z)}{3} \left[\sum_{k=1}^{3} H_2(W_3^k z^{1/3}) G_2(W_3^k z^{1/3}) \right]$$

[5/8]

Solution?

The above equalities lead to the following PR conditions:

$$\sum_{k=1}^{3} H_i(W_3^k z^{1/3}) G_i(W_3^k z^{1/3}) = 3, \text{ for } i = 0, 1, 2$$

and

$$\sum_{k=1}^{3} H_j(W_3^k z^{1/3}) G_i(W_3^k z^{1/3}) = 0, \text{ for } i \neq j \text{ and } i = 0, 1, 2; j = 0, 1, 2$$

[8/8]

ii. If we assume that $H_i(z) = G_i(z^{-1})$ for $i = 0, 1, 2$ then the PR conditions become

$$\sum_{k=1}^{3} G_i(W_3^k z^{1/3}) G_i(W_3^k z^{-1/3}) = 3, \text{ for } i = 0, 1, 2$$

and

$$\sum_{k=1}^{3} G_j(W_3^k z^{1/3}) G_i(W_3^k z^{-1/3}) = 0, \text{ for } i \neq j \text{ and } i = 0, 1, 2; j = 0, 1, 2$$

By using the result in part (a), we conclude that the PR conditions in time domain are:

$$\langle g_i[n], g_i[n - 3k] \rangle = \delta_k, \text{ for } i = 0, 1, 2$$

and

$$\langle g_j[n], g_i[n - 3k] \rangle = 0, \text{ for } i \neq j \text{ and } i = 0, 1, 2; j = 0, 1, 2$$

iii. We first note that the $g_0[n]$ provided satisfies

$$\langle g_0[n], g_0[n - 3k] \rangle = \delta_k,$$

we need to design $g_1[n]$ and $g_2[n]$ such that

$$\langle g_i[n], g_i[n - 3k] \rangle = \delta_k, \text{ for } i = 1, 2$$

and

$$\langle g_j[n], g_i[n - 3k] \rangle = 0, \text{ for } i \neq j \text{ and } i = 0, 1, 2; j = 0, 1, 2$$

[3/8]

Shortest filters implies that g_1 and g_2 have both three taps, i.e., $g_1[n] = g_2[n] = 0$ for $n \neq -1, 0, 1$.
Symmetry implies that $g_1[1] = g_1[-1]$, so the PR conditions lead to the following requirements for $g_1[n]$:

$$\langle g_1[n], g_1[n - 3k] \rangle = \delta_k \Rightarrow g_1^2[0] + 2g_1^2[1] = 1$$

and

$$\langle g_0[n], g_1[n - 3k \rangle = 0 \Rightarrow g_1[0]/\sqrt{3} + 2g_1[1]/\sqrt{3} = 0.$$

This yields (solution unique up to a sign): $g_1[0] = 2/\sqrt{6}$ and $g_1[1] = g_1[-1] = -1/\sqrt{6}$

[6/8]

Anti-symmetry implies $g_2[1] = -g_2[-1]$ and PR conditions yield $g_2[0] = 0$, $g_2[1] = 1/\sqrt{2}$ and $g_2[-1] = -1/\sqrt{2}$.

[8/8]

SOLUTIONS

2. (a) We first note that $G_1(z)G_1(z^{-1}) + G_1(-z)G_1(-z^{-1}) = 2$ which confirms that it is possible to design an orthogonal PR filter-bank. Therefore using shift and modulation we obtain:

$$\begin{aligned} G_0(z) &= -z^{-1}G_1(-z^{-1}) \\ &= -\tfrac{\sqrt{2}}{4(\alpha-1)}z^{-1}\left(-z^3 + (\alpha-2)z^2 + (2\alpha-1)z + \alpha\right) \\ &= \tfrac{\sqrt{2}}{4(\alpha-1)}\left(z^2 - (\alpha-2)z - (2\alpha-1) - \alpha z^{-1}\right) \end{aligned}$$

[4/8]

The other two filters are:

$$H_0(z) = G_0(z^{-1}) = \frac{\sqrt{2}}{4(\alpha-1)}\left(z^{-2} - (\alpha-2)z^{-1} - (2\alpha-1) - \alpha z^{1}\right)$$

$$H_1(z) = G_1(z^{-1}) = \frac{\sqrt{2}}{4(\alpha-1)}\left(z^{3} + (\alpha-2)z^{2} - (2\alpha-1)z^{1} + \alpha\right)$$

[8/8]

(b) We immediately see that $z_1 = 1$ is a zero for $G_1(z)$, so $(z^{-1} - 1)$ is a factor and this leads to this factorization

$$G_1(z) = \frac{\sqrt{2}}{4(\alpha-1)}(z^{-1} - 1)(z^{-2} + (\alpha-1)z^{-1} - \alpha)$$

[3/8]

The remaining two roots are

$$z_{2,3} = \frac{-(\alpha-1) \pm \sqrt{(\alpha-1)^2 + 4\alpha}}{2}$$

which yields $z_2 = 1$ and $z_3 = -\alpha$.

[8/8]

(c) $G_1(z)$ has two zeros at $z = 1$. Since $H_1(z) = G_1(z^{-1})$, then $H_1(z)$ has also two roots at $z = 1$, this means that $H_1(z)$ annihilates discrete-time polynomials up to degree one. That is, any sequence of the form $x[n] = an + b$ for some constants a, b is annihilated by $H_1(z)$. Thus, if we ignore border effects, then $y_1[n] = 0$ and $y_2[n] = 0$ but $y_0[n] \neq 0$.

(d) In this case all signals are non-zero because we require three roots at $z = 1$ to annihilate quadratic polynomials.

SOLUTIONS

3. (a) Since the functions $\varphi(t)$, $\psi(t)$, $\sqrt{2}\psi(2t)$ and $\sqrt{2}\psi(2t-1)$ are orthonormal, we don't need to find the dual basis: $\tilde{\varphi}_i(t) = \varphi_i(t)$. [3/7]

Consequently

$$c_1 = \langle f(t), \varphi(t) \rangle = 2\int_0^{1/2} t dt + \int_{1/2}^{1} dt = \frac{3}{4}$$

$$c_2 = \langle f(t), \psi(t) \rangle = 2\int_0^{1/2} t dt - \int_{1/2}^{1} dt = -\frac{1}{4}$$

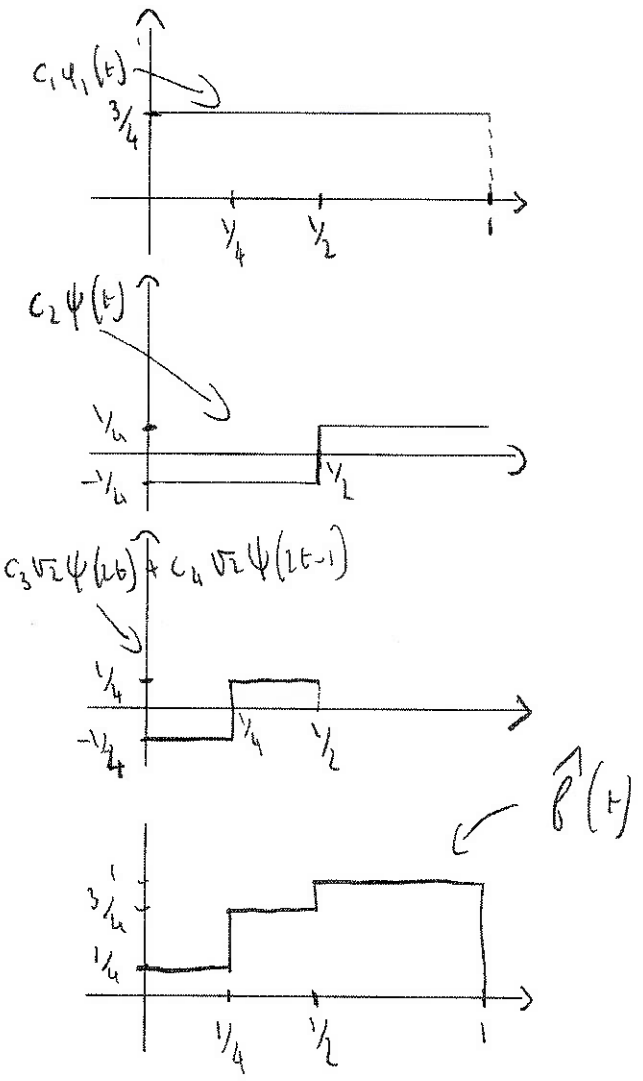
$$c_3 = \langle f(t), \sqrt{2}\psi(2t) \rangle = 2\sqrt{2}\left(\int_0^{1/4} t dt - \int_{1/4}^{1/2} t dt\right) = -\frac{\sqrt{2}}{8}$$

$$c_4 = \langle f(t), \sqrt{2}\psi(2t-1) \rangle = 0$$

[7/7]

SOLUTIONS

(b) Solution sketched below:



SOLUTIONS

(c)

$$\|\epsilon(t)\|^2 = 2\int_0^{1/4}(2t-1/4)^2dt = 1/96$$

(d) Using Parseval identity we have that

$$\|\hat{f}(t)\|^2 = \sum_{i=1}^{4} c_i^2 = 9/16 + 1/16 + 1/32 = 21/32.$$

Moreover,

$$\|f(t)\|^2 = 4\int_0^{1/2} t^2dt + \int_{1/2}^{1} dt = 1/6 + 1/2 = 2/3.$$

Therefore

$$\|\epsilon(t)\|^2 + \|\hat{f}(t)\|^2 = 1/96 + 21/32 = 2/3 = \|f(t)\|^2.$$

Solutions

4. (a) Using Poisson summation formula we obtain:

$$\sum_k \varphi(t-k) = \sum_n \hat{\varphi}(2\pi n) e^{-j2\pi nt}$$

then $\sum_k \varphi(t-k) = 1$ implies $\sum_n \hat{\varphi}(2\pi n) e^{-j2\pi nt}$ and this last equality is satisfied when $\hat{\varphi}(2\pi n) = 1$ for $n = 0$ and $\hat{\varphi}(2\pi n) = 0$ for $n \neq 0$.

(b) We first note that, since $\hat{\varphi}(2\pi n) = 1$ for $n = 0$ and $\hat{\varphi}(2\pi n) = 0$ for $n \neq 0$ then

$$\sum_k \varphi(t-k) = 1.$$

[2/9]

We then use the moment property which says that:

$$\hat{\varphi}^{(1)}(\omega) = -j \int_{-\infty}^{\infty} t\varphi(t) e^{-j\omega t} dt.$$

as follows:
Take the periodic signal

$$\sum_k (t-k)\varphi(t-k)$$

and using the Poisson summation formula, we obtain:

$$\sum_k (t-k)\varphi(t-k) = j \sum_n \hat{\varphi}^{(1)}(2\pi n) e^{-j2\pi nt},$$

where we have also used the moment property of the Fourier transform. [5/9]

Since $\hat{\varphi}^{(1)}(2\pi n) = 0$ for $n \neq 0$ then

$$\sum_k (t-k)\varphi(t-k) = j\hat{\varphi}^{(1)}(0).$$

Therefore

$$t - \sum_k k\varphi(t-k) = j\hat{\varphi}^{(1)}(0),$$

where we have used the fact that $\sum_k \varphi(t-k) = 1$. By rearranging the terms we obtain:

$$\sum_k k\varphi(t-k) + j\hat{\varphi}^{(1)}(0) = t,$$

and by using again the fact that $\sum_k \varphi(t-k) = 1$, we obtain

$$\sum_k \left(k + j\hat{\varphi}^{(1)}(0)\right) \varphi(t-k) = t.$$

This allows us to claim that

$$\sum_k c_k \varphi(t-k) = t.$$

for

$$c_k = k + j\hat{\varphi}^{(1)}(0).$$

[9/9]

SOLUTIONS

(c) We want

$$\sum_k c_k \varphi(t-k) = e^{j\omega_0 t}.$$

We set $c_k = c_0 e^{j\omega_0 k}$ and obtain

$$\sum_k c_0 e^{j\omega_0 k} \varphi(t-k) = e^{j\omega_0 t}.$$

[2/10]

We then divide both side of the equation by $e^{j\omega_0 t}$ and obtain

$$\sum_k c_0 e^{-j\omega_0 (t-k)} \varphi(t-k) = 1.$$

The quantity

$$\sum_k e^{-j\omega_0 (t-k)} \varphi(t-k)$$

is a period function of period $T = 1$.

[5/10]

We can therefore apply Poisson sum formula which yields

$$c_0 \sum_k e^{-j\omega_0 (t-k)} \varphi(t-k) = c_0 \sum_n \hat{\varphi}(\omega_0 + 2\pi n) e^{j2\pi n t} = 1,$$

where we have used the fact that the Fourier transform of $e^{-j\omega_0 t}\varphi(t)$ is $\hat{\varphi}(\omega + \omega_0)$. The equality

$$c_0 \sum_n \hat{\varphi}(\omega_0 + 2\pi n) e^{j2\pi n t} = 1,$$

is then satisfied when $\hat{\varphi}(\omega_0 + 2\pi n) = 0$ for $n \neq 0$ and $\hat{\varphi}(\omega_0) = 1/c_0$.

[10/10]